Energy Storage Technology (ME4121)

Open Elective (For other engineering department students, not for M.E.)

Contact Hours : 3-0-0 Credit:3 (Marks:100)

Syllabus:
Need for energy storage, Electrical energy storage system- Battery types and parameters, Lead Acid battery cycle VLRA Battery, Battery charging, Lithium ion battery, Mathematical modeling of lead acid battery, Designer's choice of battery, ultra and super capacitor. Mechanical energy storage device-flywheel, compressed air storage system. Pump storage plants. Hydrogen storage and fuel cell system. Fuel cell types and thermodynamics., efficiency and voltage. Chemical Energy storage- Sensible heat storage system, Latent heat storage system, Thermo-chemical energy storage. Energy storage in phase change material.

Topics	No. of periods
Need for energy storage, Electrochemical energy storage system and Battery, classification and different types: Metal air, Lead Acid, Nickel Cadmium, Sodium Sulphur and Lithium ion battery; Battery characteristics and performance parameters, Secondary battery; charging-discharging cycle, Designer's choice of battery, Electrical energy storage and capacitor: ultra and super capacitors	09
Super conducting Magnetic Energy Storage (SMES)	02
Mechanical energy storage device- flywheel, compressed air storage system. Pump storage plants, merits and demerits.	07
Hydrogen production and storage: Different methods of hydrogen production; chemical, thermo-chemical, photochemical, electro-chemical; electrolyzer and fuel cell. Fuel cell types and thermodynamics, efficiency and voltage. Reversible fuel cell; Thermal splitting of water	08
Other chemical Energy storage- Bio-fuels and Hydrated Salts, Accumulators with internal and external storage. Graphene based composites for electrochemical energy storage	05
Thermal Energy Storage System- Sensible heat storage system, Latent heat storage system, Thermo-chemical energy storage. Concept of solar Pond. Basics of Thermo-electric generator. Heat Pipes and Vapour Chambers.	05
Economics of the Energy Storage, Special considerations for automotive and traction applications	06
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Text Books:

1. Energy Storage Systems by David Elliott, IOP Publishing Ltd. Bristol, UK, 2017. ISBN 978-0-7503-1531-9 (ebook).

2. Storing Energy Ist Edition by Trevor Letcher, Imprint Elsevier, 2016, Hardcover ISBN: 9780128034408,eBook ISBN: 9780128034491.

3. Solar Energy: Principles of Thermal Collection and Storage by S. P. Sukhatme and J.K.Nayak, Tata Mc Graw-Hill Publishing Company Limited.